The worker/harness weld is the regular bicomodule: deciding Conjecture 5.1

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Abstract

Let S, T be sets and let $\Delta S = S \cdot y^S$ denote the store (costate) comonad on S — equivalently, the indiscrete category on S and a comonoid in $(\mathrm{Poly}, \triangleleft, y)$. The worker/harness weld polynomial is $B = \Delta S \triangleleft \Delta T$ and the worker grading is $A = \Delta S \otimes \Delta T = \Delta(S \times T)$. Conjecture 5.1 of the workers/bicomodules note asserts three things: **(1)** B carries a genuine bicomodule structure (a horizontal morphism of $\mathrm{Cat}^\sharp = \mathrm{Comod}(\mathrm{Poly})$, i.e. a parametric right adjoint $\Delta T\text{-Set} \to \Delta S\text{-Set}$); **(2)** under the SGF slice factorization the worker A is the right-adjoint *content* factor; and **(3)** the data discarded by the workers retract idempotent e is the coproduct Σ-aggregation tail. We decide all three. Clause (1) is **TRUE**: B is the *regular* $(\Delta T, \Delta S)$-bicomodule, with left coaction $\lambda = \delta_{\Delta S} \triangleleft \Delta T$ and right coaction $\rho = \Delta S \triangleleft \delta_{\Delta T}$ (Theorem 4), the five bicomodule axioms being machine-checked as equalities of natural transformations on five $(|S|, |T|)$ shapes. Clause (2) is **FALSE**: computing the represented prafunctor gives $\bar{F}_B(V) = (T \times V)^S$, whose SGF content factor is the *power* $(-)^S$, not A (Theorem 6). Clause (3) is **TRUE**: the Σ-aggregation tail is indexed by the branch-map set $T^S = \{g \colon S \to T\}$, exactly the data e forgets (Theorem 7). The note's "Gap 5.2" (an apparent reversal of adjointness) is dissolved: T^S is the *index* of a left-adjoint Σ, not right-adjoint exponential content. The honest summary: the weld is a bicomodule and its retract's discarded branch data is the SGF aggregation tail — but state-entanglement (A) and data-aggregation (the Σ-tail) are genuinely *different* collapses. We do not inflate this partial bridge into a full worker = content identification; clause (2) is a refutation.

1 Statement decided

Throughout, $\mathrm{Poly} \subset [\mathrm{Set}, \mathrm{Set}]$ is the category of polynomial functors, fully faithful into $[\mathrm{Set}, \mathrm{Set}]$: a polynomial $p = \sum_{s \in S_p} y^{p[s]}$ is the functor $p(X) = \sum_s X^{p[s]}$, and a morphism of polynomials *is* a natural transformation. The composition product $\triangleleft$ is functor composition, $(p \triangleleft q)(X) = p(q(X))$, with unit y; the Dirichlet product $\otimes$ has position set $S_p \times S_q$ and directions $p[s] \times q[t]$, unit y. A comonoid in $(\mathrm{Poly}, \triangleleft, y)$ is a small category (Ahman–Uustalu; SGF Prop. 5.2).

The **store comonad** ΔS (SGF Ex. 5.6) has position set S and directions S at every position; as a comonad $\Delta S(W) = S \times W^S$, with $\delta_S(s,\varphi) = (s,\, i \mapsto (i,\varphi))$ and $\varepsilon_S(s,\varphi) = \varphi(s)$. As a category it is the **indiscrete groupoid** on S (a unique isomorphism between any two objects).

Definition 1 (the weld and the worker)**.** Fix sets S, T. The *weld* polynomial is $B := \Delta S \triangleleft \Delta T$. The *worker* grading is $A := \Delta S \otimes \Delta T = \Delta(S \times T)$.

Conjecture 5.1 of `expository/2026-09-20-workers-bicomodules-pra.tex` asserts:

(1) *(existence)* B carries a (c,d)-bicomodule structure for some comonoids c, d — a horizontal morphism of $\mathrm{Cat}^\sharp = \mathrm{Comod}(\mathrm{Poly})$, i.e. a parametric right adjoint (PRA) functor c-Set $\to$ d-Set by the SGF correspondence (SGF Thm. 4.28 / Prop. 3.9);

(2) *(identification)* under the SGF slice factorization (Prop. 3.6) of the represented prafunctor, the worker A is the right-adjoint *content* factor $F/1$;

(3) *(tail)* the data discarded by the workers retract idempotent e (the non-constant branch maps $g\colon S \to T$) is the coproduct Σ-aggregation tail.

Theorem 2 (verdict)**.** *With $c = \Delta T$, $d = \Delta S$:*

clause	***verdict***	***content***
(1) existence	***TRUE***	*B is the regular $(\Delta T, \Delta S)$-bicomodule; general comonoid nonsense (Lemma 3, Theorem 4).*
(2) worker = content	***FALSE***	*the content factor is the power $(-)^S$, not A (Theorem 6).*
(3) discarded = Σ-tail	***TRUE***	*the Σ-tail is $\Sigma_{g \in T^S}$, indexed by the branch maps (Theorem 7).*

In addition, the note's Gap 5.2 *("apparent reversal of adjointness") is* dissolved *(Theorem 8): the branch-map set T^S is the indexing set of a genuine left adjoint Σ, not right-adjoint exponential content.*

The honest one-line summary: *the weld is a bicomodule, and the retract's discarded branch data is exactly the SGF aggregation tail — but the worker A is a different collapse from the bicomodule content.* State-entanglement $\neq$ data-aggregation.

2 Clause (1): the weld is the regular bicomodule — TRUE

Recall (SGF Def. 4.2, Leibniz order of Rmk. 4.3) that a (c,d)-**bicomodule** $m\colon c \to d$ is a triple (m, λ, ρ) with a left d-coaction $\lambda\colon m \to d \triangleleft m$ and a right c-coaction $\rho\colon m \to m \triangleleft c$ satisfying

(34) left counit $(\varepsilon_d \triangleleft m) \circ \lambda = \mathrm{id}$; left coassociativity $(\delta_d \triangleleft m) \circ \lambda = (d \triangleleft \lambda) \circ \lambda$;

(35) right counit $(m \triangleleft \varepsilon_c) \circ \rho = \mathrm{id}$; right coassociativity $(\rho \triangleleft c) \circ \rho = (m \triangleleft \delta_c) \circ \rho$;

(36) compatibility $(\lambda \triangleleft c) \circ \rho = (d \triangleleft \rho) \circ \lambda$.

By SGF Thm. 4.28 / Prop. 3.9, $\mathrm{Comod}(\mathrm{Poly})(c,d) \simeq \mathrm{PRA}(c\text{-Set}, d\text{-Set})$: a bicomodule $m\colon c \to d$ *is* a parametric right adjoint c-Set $\to$ d-Set.

Lemma 3 (regular bicomodule)**.** *For any comonoids $c = (c, \varepsilon_c, \delta_c)$ and $d = (d, \varepsilon_d, \delta_d)$ in a monoidal category $(V, \triangleleft, y)$, the object $d \triangleleft c$ is a (c, d)-bicomodule with*

$$\lambda := \delta_d \triangleleft c \colon d \triangleleft c \to d \triangleleft d \triangleleft c = d \triangleleft (d \triangleleft c), \qquad \rho := d \triangleleft \delta_c \colon d \triangleleft c \to d \triangleleft c \triangleleft c = (d \triangleleft c) \triangleleft c.$$

Proof. Axioms (34)/(35): λ is δ_d whiskered on the right by c, so the left-counit and left-coassociativity laws are the comonoid laws $(\varepsilon_d \triangleleft d) \circ \delta_d = \mathrm{id}$ and $(\delta_d \triangleleft d) \circ \delta_d = (d \triangleleft \delta_d) \circ \delta_d$ of d, whiskered by $\triangleleft c$; likewise (35) is the comonoid laws of c whiskered by $d\triangleleft$. Axiom (36): λ acts on the first (d) tensor factor and ρ on the last (c) factor; since $\triangleleft$ is a bifunctor, the two whiskerings commute (interchange law), and both $(\lambda \triangleleft c) \circ \rho$ and $(d \triangleleft \rho) \circ \lambda$ equal the single map $\delta_d \triangleleft \delta_c \colon d \triangleleft c \to d \triangleleft d \triangleleft c \triangleleft c$. □

Lemma 3 is standard — the comonoid $\triangleleft$-product of two comonoids is the regular bicomodule between them. Its consequence here is the point.

Theorem 4 (clause (1))**.** *The weld $B = \Delta S \triangleleft \Delta T$ carries the regular $(\Delta T, \Delta S)$-bicomodule structure ($c = \Delta T$, $d = \Delta S$, carrier $d \triangleleft c = \Delta S \triangleleft \Delta T$), with $\lambda = \delta_{\Delta S} \triangleleft \Delta T$ and $\rho = \Delta S \triangleleft \delta_{\Delta T}$. Hence B is a horizontal morphism of* $\mathrm{Cat}^\sharp$*, i.e. a prafunctor ΔT-Set $\to \Delta S$-Set.* Conjecture 5.1 clause (1): TRUE.

Proof. Immediate from Lemma 3 with $c = \Delta T$, $d = \Delta S$; both ΔS and ΔT are comonoids (store comonads). □

Machine check. All five axioms — (34) counit and coassociativity, (35) counit and coassociativity, (36) compatibility — were verified as equalities of natural transformations (Godement calculus on nested store-tuples) for $(|S|, |T|) \in \{(2,2), (2,3), (3,2), (1,2), (2,1)\}$ on test objects of size 2 and 3; every case returned `True`. Script `scratch/2026-10-16-weld-bicomodule-verify.py`; registry node `bicomodule-axioms-machine-checked` **[computed]**. This is independent corroboration of Theorem 4, not a dependency: the theorem stands on the proved Lemma 3.

Which two comonoids? This dissolves the note's worry ("Gap 5.3") that $\Delta S \triangleleft \Delta T$ is a *monoidal product* of two comonoids rather than, *a priori*, a bicomodule *between* two fixed ones. The regular-bicomodule construction turns the $\triangleleft$-product of *any* two comonoids into a bicomodule between them; the pair is $(c, d) = (\Delta T, \Delta S)$.

3 The represented prafunctor

Because ΔS is the indiscrete groupoid on S, it is equivalent as a category to the terminal category $\mathbf{1}$ (for $S \neq \varnothing$: a unique isomorphism between any two objects). Hence ΔS-Set $\simeq$ Set: a copresheaf on ΔS is a single set with canonical identifications, and $\lim X \cong X(s_0) \cong \mathrm{colim}\, X$ for every object s_0. Write $E_c \colon c\text{-Set} \simeq \mathrm{Set}$ for this equivalence ($E_c^{-1} V$ is the constant copresheaf V). We transport the prafunctor of B to an endofunctor $\bar{F}_B$ of Set; right-adjointness and the SGF factorization are equivalence-invariant.

Theorem 5 (represented prafunctor)**.** $\bar{F}_B(V) \cong (T \times V)^S = T^S \times V^S$*. As a polynomial endofunctor of* Set*, $\bar{F}_B = T^S \cdot y^S$ (positions T^S, directions S). Concretely*

$$\bar{F}_B(X) = \{(g, h) \colon g \colon S \to T,\ h \colon S \to X\}.$$

Proof. The regular bicomodule $d \triangleleft c$ is the composite over the unit y of two one-sided comodules:

$$\Delta T \xrightarrow{G:=(\Delta T \to y)} y = \mathrm{Set} \xrightarrow{H:=(y \to \Delta S)} \Delta S, \qquad \text{carrier } H \triangleleft G = \Delta S \triangleleft \Delta T = B,$$

where G is the right ΔT-comodule with carrier ΔT and $\rho = \delta_{\Delta T}$, and H is the left ΔS-comodule with carrier ΔS and $\lambda = \delta_{\Delta S}$. Bicomodule composition over y is just $\triangleleft$ (no equalizer, since y is the unit), and $H \triangleleft G = \Delta S \triangleleft \Delta T = B$. Prafunctors compose, so $F_B = F_H \circ F_G$.

Factor $F_G \colon \Delta T\text{-Set} \to \mathrm{Set}$. As a ΔT-variable polynomial (SGF Prop. 3.6, Set[c]), the right ΔT-comodule with carrier ΔT is $F_G(X) = \sum_{t \in T} \Delta T\text{-Set}(y_t, X) = \sum_{t \in T} X(t)$, where $y_t = \Delta T(t, -)$ is the representable at t. In the indiscrete category y_t is the terminal copresheaf $\mathbf{1}$ (as $\Delta T(t, t') = 1$), so $\Delta T\text{-Set}(y_t, X) = \lim X$. Hence $F_G(X) = \sum_{t \in T} \lim X$, and after transport $\bar{G}(V) = T \times V$.

Factor $F_H \colon \mathrm{Set} \to \Delta S\text{-Set}$. A left ΔS-comodule is a functor $\Delta S \to \mathrm{Poly}$ (SGF Prop. 5.7); the regular one (carrier ΔS, $\lambda = \delta_{\Delta S}$) assigns to each object $a \in S$ the component polynomial $(\Delta S)_a = y^S$ (one position over a, with S directions — read off δ_S). So $F_H(W)(a) = (y^S)(W) = W^S$ (constant copresheaf), and after transport $\bar{H}(W) = W^S$.

Therefore $\bar{F}_B(V) = \bar{H}(\bar{G}(V)) = (T \times V)^S = T^S \times V^S$, and $(T \times V)^S = \{S \to T \times V\} = \{(g, h) \colon g \colon S \to T,\ h \colon S \to V\}$. □

The point of the whole problem. $\bar{F}_B = T^S\, y^S$ is *neither* the underlying polynomial of $A = (S \times T)\, y^{S \times T}$ *nor* that of $B = (S \cdot T^S)\, y^{S \times T}$. The represented prafunctor of a bicomodule is *not* the underlying polynomial functor of its carrier — exactly the trap the note flagged. Machine-checked cardinalities at $|S| = |T| = 2$, $V = 2$: $\bar{F}_B = 16$, $A_{\text{under}} = 64$, $B_{\text{under}} = 128$.

4 The SGF factorization, and clauses (2) and (3)

Apply the SGF slice factorization (Prop. 3.6) to $\bar{F}_B(V) = \sum_{g \in T^S} V^S$:

- $\bar{F}_B(\mathbf{1}) = T^S$;
- **content** $\bar{F}_B/\mathbf{1} \colon \mathrm{Set} \to \mathrm{Set}^{T^S}$, $V \mapsto (V^S)_{g \in T^S} = \Delta_{T^S} \circ (-)^S$. Both the power $(-)^S$ and the diagonal Δ_{T^S} are right adjoints, so $\bar{F}_B/\mathbf{1}$ is a right adjoint; the genuine connected-limit content is the **power** $(-)^S$;
- **Σ-tail** $\Sigma \colon \mathrm{Set}^{T^S} \to \mathrm{Set}$, the coproduct over $T^S = \{g \colon S \to T\}$.

Theorem 6 (clause (2): FALSE)**.** *The right-adjoint content factor of the weld prafunctor is the power $(-)^S$, not the worker A. Indeed $A = \Delta(S \times T)$ has represented functor $V \mapsto V^{S \times T}$ (exponent $S \times T$), whereas the content has exponent S; these differ whenever $|T| \geq 2$. No handedness convention repairs this: the opposite reading gives content $(-)^T$, still a power, never A.*

Proof. The content exponent is S (the directions of $\bar{F}_B = T^S y^S$); A's represented functor has exponent $S \times T$ by the computation of Theorem 5 applied to $A = \Delta(S \times T)$, giving $\bar{A}(V) = V^{S \times T}$. For $|T| \geq 2$ we have $|S| < |S \times T|$, so the two functors differ on any V with $|V| \geq 2$. □

Theorem 7 (clause (3): TRUE)**.** *The SGF Σ-aggregation tail is the coproduct $\Sigma_{g \in T^S}$ indexed by the full branch-map set $T^S = \{g \colon S \to T\}$ — precisely the "non-constant branch maps" that the retract idempotent e of the workers theorem forgets.*

Proof. The tail index is $\bar{F}_B(\mathbf{1}) = T^S$ by Theorem 5; $T^S = \mathrm{Hom}(S, T)$ is the set of branch maps, and $e(s, g) = (s, \mathrm{const}_{g(s)})$ collapses each g to a constant, forgetting exactly the non-constant elements of T^S. □

Theorem 8 (Gap 5.2 dissolved)**.** *There is no adjointness reversal. The branch maps T^S appear as the* indexing set *of the left-adjoint coproduct Σ_{T^S}, not as right-adjoint exponential content. "T^S is an exponential object" is a statement about the* set*, not about the adjointness of $\Sigma_{T^S} : \mathrm{Set}^{T^S} \to \mathrm{Set}$, which is left adjoint to the diagonal. The genuinely right-adjoint (exponential / connected-limit) content $(-)^S$ is* kept *in $F/\mathbf{1}$, exactly as SGF predicts.*

Proof. $\Sigma_{T^S} \dashv \Delta_{T^S}$, so Σ_{T^S} is a left adjoint for any index set, exponential or not; and $(-)^S$ is right adjoint to the copower $S \times (-)$, hence connected-limit-preserving content. The note's Gap 5.2 conflated the two. □

5 The precise role of the worker

The worker A is not the content, but it has an exact location inside the weld prafunctor. The section $\sigma\colon A \to B$, $\sigma_{\mathbf{1}}(s,t) = (s, \mathrm{const}_t)$, embeds A's T-coordinate as the *constant* branch maps $T \hookrightarrow T^S$. In the factorization of $\bar{F}_B$, the current input state S lives in the content exponent and the branch map lives in the Σ-tail index T^S; so

$$A\text{'s positions } S \times T \;=\; (\text{content-input } S) \times (\text{constant-branch slice } T \subseteq T^S \text{ of the } \Sigma\text{-tail index}).$$

The worker is the constant/diagonal slice of the *aggregation tail* — a statement about Σ, not about $F/\mathbf{1}$. This sharpens, rather than rescues, the negative verdict of clause (2): entanglement selects constant branches; it does not supply the bicomodule content.

6 Grant placement

The weld now has a precise home in Poly: it *is* a data-migration prafunctor ΔT-Set $\to \Delta S$-Set, $X \mapsto (T \times X)^S$ — "read the whole input state S; return, for each input, an output token in T and a value in X," with the branch-map choice aggregated by Σ. This places the *state axis* of the compositional-correctness program inside the same framed bicategory $\mathrm{Cat}^\sharp$ that houses the directed (retrofunctor) axis — the positive deliverable the note hoped for. But the identification "worker = PRA content" is *false*: state-entanglement (A) and data-aggregation (the Σ-tail) are genuinely different collapses, and the agent-orchestration weld semantics must keep them distinct.

7 Scope and honesty

- Lemma 3 is standard category theory; the contribution is clauses (2)/(3), the explicit $\bar{F}_B = (T \times V)^S$ with its factorization, and the dissolution of Gap 5.2.
- Theorem 5 uses the two single-sided represented-functor formulas (F_G, F_H); these are the SGF $\mathrm{Set}[c]$ / Prop. 5.7 dictionaries applied to the indiscrete case, where c-Set $\simeq$ Set, representables are terminal, and component polynomials are y^S. Each step is elementary but was *derived*, not re-brute-forced from the bicomodule-composition equalizer; the axiom check (§2) and the cardinality check (§3) are the computational corroboration.
- This note decides exactly the three clauses of Conjecture 5.1 plus Gap 5.2; no claim is made beyond them. In particular clause (2) is a *refutation*, and the partial bridge (clauses 1 and 3) is not inflated into a full worker = content identification.